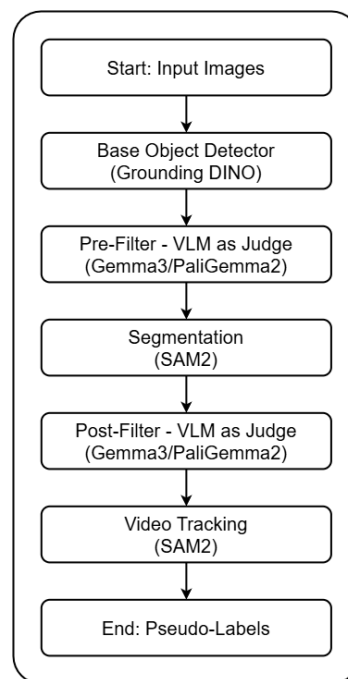


Towards Robust Unsupervised Object Detection

A Multi-Layer Framework Using Open-Vocabulary Zero-Shot Detector, Vision-Language Models and Segmentation for Data Annotation



Devashish Singha Roy

Master Programme in Applied Artificial Intelligence
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Luleå University of Technology
Department of Computer Science, Electrical and Space Engineering

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Abstract

Object detection has witnessed significant progress in recent years; however, its application in the Nordic region has remained limited. This is largely due to region's unique geographical and environmental characteristics which creates a significant domain gap between available aerial datasets and the conditions existing in Nordic regions. Additionally, there is notable scarcity of annotated data in this region, which further limits the industrial adoption of current object detection methods.

This master's thesis addresses these challenges by proposing a multi-layered framework that leverages state-of-the-art pre-trained models such as Grounding DINO, SAM2, and Vision-Language Models such as Gemma3 and PaliGemma2, to generate high-quality pseudo-labels from unannotated data in an unsupervised manner. These pseudo-labels can then be used for training, evaluation, and fine-tuning of object detection models, ultimately supporting the development of robust and scalable object detection solutions.

Keywords: Artificial Intelligence, Deep Learning, Computer Vision, Unsupervised Learning, Pseudo-labeling, Zero-shot Open-vocabulary Object Detection, Grounding DINO, Vision-Language Models, Gemma3, PaliGemma2, Semantic Filtering, Prompt Engineering, Segmentation, SAM2, Aerial Imagery, Video Tracking, Multi-layer Pipeline, Scalable Annotation

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Chapter 1

Introduction

Object detection is a fundamental task in computer vision with broad applications in surveillance, autonomous navigation, and infrastructure monitoring. In the aerial object detection domain, unmanned aerial vehicles (UAVs) have been extensively used for search and rescue operations [1], wildlife monitoring [2], surveillance [3], and scene understanding [4, 5]. While a significant progress have been made in recent years, much of it relied on large scale annotated datasets from well-studied regions with standard environmental conditions. These datasets rarely cover the unique geographical features of Nordic region, such as year-long snow coverage, seasonal lighting variations and sparsely populated landscapes. This gives rise to significant domain gap between commonly used datasets and the actual conditions encountered in Nordic aerial imagery. To fill this gap, Nordic Vehicle Dataset (NVD) was introduced, providing a valuable benchmark on several object detection tasks [6]. While NVD provides a strong foundation for research in Nordic region, manual data collection and annotation remains time consuming and costly especially when scaling to larger datasets, adapting to new object categories, or accounting for seasonal variations. As a result, there is a growing need of self-supervised or unsupervised labeling approaches that can reduce dependency on manual collection and annotation of data while been able to maintain semantic accuracy.

1.1 Traditional techniques used in object detection

Traditional object detection techniques initially relied on hand-crafted features and classical machine learning algorithms. Notable among these was the Viola-Jones detector [7], which used Haar-like features and a cascade of classifiers for real-time face detection. This was followed by approaches leveraging Histogram of Oriented Gradients (HOG) features combined with Support Vector Machines (SVMs), such as the method proposed by Dalal and Triggs [8]. While these techniques performed well in constrained scenarios, they struggled with variations in object scale, occlusion, and background clutter.

The introduction of Convolutional Neural Networks (CNNs) revolutionized object detection. R-CNN [9] proposed using CNNs to extract features from region proposals, followed by classification. Subsequent improvements such as Fast R-CNN [10] and Faster R-CNN [11] integrated region proposal generation and feature extraction into a unified framework, significantly improving speed and accuracy. Parallel to this, single-stage detectors like YOLO (You Only Look Once) [12] and SSD (Single Shot MultiBox Detector) [13] provided real-time detection by directly predicting object classes and bounding boxes from feature maps without using region proposals.

Despite their success, these supervised methods require large volumes of annotated training data and are constrained to closed-set vocabularies. This makes them less effective in open-world settings or domains with limited labeled data. These limitations have led to a shift towards weakly-supervised, self-supervised, unsupervised, and vision-language driven object detection techniques, which aim to reduce annotation effort while enabling broader generalization.

1.2 Motivation

Data annotation is a time-consuming, expensive, and often domain-specific process that limits the scalability of supervised learning systems. While state-of-the-art vision-language models (VLMs) such as Gemma3 and Paligemma2 and zero-shot open-vocabulary models such as Grounding DINO offer powerful capabilities in object detection and semantic understanding, their raw outputs often suffer from issues like noise, redundancy,

and temporal inconsistency especially in video data.

This thesis is motivated by the need to build a reliable, scalable, and modular pseudo-labeling pipeline that can effectively harness these pre-trained models while overcoming their inherent limitations. By integrating these multiple state-of-the-art models across a layered architecture, combining object detection, semantic filtering, image segmentation, and video tracking, this work aims to reduce manual labeling effort in creating high-quality annotations from unlabeled image and video datasets, thereby enabling scalable and sustainable AI development.

1.3 Problem statement

The lack of annotated data remains a major bottleneck in training and evaluating deep learning models for object detection and segmentation, especially in complex and harsh environments like extreme snowy conditions. Manual annotation is not only resource-intensive but also impractical for large-scale or dynamic datasets. This limitation necessitates the development of automated methods to generate pseudo-labels without human supervision.

One promising direction is the use of zero-shot, open-vocabulary detector such as Grounding DINO, vision-language models like PaliGemma2, Gemma3 for noise filtration and SAM2 for segmentation. However, these models face several critical challenges:

1. **Small Object Detection Deficiency:** Preliminary experiments revealed that Grounding DINO fails to reliably detect small and very small objects, particularly in snowy environments where object boundaries are ambiguous.
2. **Sensitivity to Thresholds and Prompts:** The performance of Grounding DINO is highly sensitive to textual prompts and threshold settings (both box and text), making the system unstable and difficult to generalize.
3. **Unclear Effectiveness of Segmentation in Post-Processing:** While SAM2 provides segmentation capabilities, its effectiveness in refining detection outputs through post-filtering mechanisms needs rigorous evaluation.

4. Unclear Effectiveness of Vision-Language Models as Semantic Judges: The use of vision-language models such as Gemma3 and PaliGemma2 as semantic filters to judge detection quality is a promising yet underexplored direction. Their ability to guide pseudo-labeling through prompt based reasoning raises important questions about prompt design, reliability, and filtering effectiveness.

These challenges highlight a critical gap in building a reliable, fully unsupervised labeling framework that can operate effectively in adverse conditions. This thesis aims to address these gaps through a structured exploration and evaluation of modern detection, segmentation, and vision-language models.

1.4 Objectives

The objective of this thesis is to develop and evaluate an unsupervised pseudo-labeling framework for aerial object detection by integrating recent advances in zero-shot, open-vocabulary detection, segmentation, and vision-language reasoning. This foundational objective enables subsequent investigations into specific components and configurations of the framework. The objectives are structured as follows:

1. The primary objective of this thesis is to design, implement, and evaluate an unsupervised pseudo-labeling framework for aerial object detection, leveraging zero-shot, open-vocabulary detectors, vision-language models (VLMs) and segmentation-based techniques.
2. Investigate the role of vision-language models (e.g., Gemma3, PaliGemma2) in improving the semantic filtering of noisy detections through prompt based reasoning.
3. Analyze the effect of Grounding DINO configurations, particularly box threshold and text threshold settings, on detection performance and pipeline efficiency.
4. Explore spatial (image windowing) and temporal (SAM2 video propagation) techniques to enhance detection of small or contextually subtle objects in aerial imagery.

5. Assess prompt sensitivity and linguistic formulation in both open-vocabulary zero-shot object detection (Grounding DINO) and VLM-based semantic filtering (PaliGemma2) to understand how different prompt types (affirmative, positive/negative, contextual) affect detection and filtering performance.

Through these objectives, the thesis seeks to advance practical methods for scalable, low-cost annotation in aerial imagery while contributing to sustainable and efficient AI deployment.

1.5 Sustainability

The availability of labeled data remains a major challenge in the development and deployment of machine learning systems. Manual annotation is both costly and time-consuming, limiting the pace of research and hindering the scalability of industrial applications. Techniques such as semi-supervised learning, few-shot learning, and zero-shot detection (e.g., Grounding DINO) promise sustainability in AI by reducing dependency on large-scale annotated datasets.

This thesis contributes to sustainable AI development in several ways. First, the proposed framework significantly reduces the need for manual annotation by generating high-quality pseudo-labels using pre-trained models in an unsupervised manner. This is particularly beneficial in geographically challenging regions like the Nordic countries, where collecting and annotating data is logistically difficult and resource-intensive. The ability to reuse unlabeled datasets for training helps accelerate the development process while minimizing human effort.

Second, the modular design of the framework enables selective use of components, such as switching between different vision-language models, allowing developers to optimize resource use based on available computational capacity and application needs.

Third, by reusing pre-trained models without any additional training or fine-tuning, the framework avoids the high energy costs typically associated with large-scale model development. This zero-training, plug-and-play approach contributes to sustainability by reducing redundant computation and minimizing the environmental footprint.

Finally, by addressing the domain gap through adaptable pseudo-labeling, the framework promotes longer-term reuse of data across seasons, geographic regions, and use cases. This supports the sustainable development of AI systems in real-world applications while reducing waste and increasing efficiency.

1.6 Ethical Considerations

The use of AI models for object detection and segmentation, particularly in aerial surveillance and public infrastructure, raises important ethical considerations. This thesis is focused on developing a unsupervised pseudo-labeling framework that reduces the need for manual annotation. While the approach improves data efficiency and scalability, it also introduces risks related to bias, privacy, and accountability.

First, pre-trained models such as Grounding DINO and vision-language models such as Gemma3 and PaliGemma2 are trained on large, generic datasets. These models may encode biases present in their training data, which could be inadvertently propagated into pseudo-labeled datasets and downstream applications. Care must be taken when applying these pseudo-labels in sensitive contexts such as traffic monitoring, surveillance, or environmental enforcement to avoid reinforcing systemic biases or inaccuracies.

Second, the use of aerial imagery, even for research purposes, may raise concerns regarding privacy, particularly in residential or urban settings. Although this thesis does not involve personal identification or individual tracking, any future deployment of the framework must ensure compliance with data protection regulations (e.g., GDPR) and local aerial data usage policies.

Finally, the automated nature of pseudo-labeling pipelines requires transparency about their limitations. False positives, overlooked objects, or errors introduced through the pipeline should be clearly communicated when using such labels for training production-level models. Ethical deployment requires continuous monitoring, human oversight where needed, and a clear understanding of the boundaries of automation.

This thesis emphasizes the responsible use of AI models and datasets and encourages further research into bias mitigation, explainability, and fairness in pseudo-labeling

frameworks.

1.7 Organization of the thesis

The thesis is organized as follows:

- Chapter 2, **Related Work**, reviews recent research in object detection, segmentation, tracking, and vision-language models relevant to this thesis work.
- Chapter 3, **Methodology**, describes the methods and techniques implemented in this study.
- Chapter 4, **Results and Discussion**, presents the outcomes of the experiments and provides an analysis of the findings.
- Chapter 5, **Conclusion**, provides a summary of the work and presents the conclusions in relation to the research objectives.

Chapter 2

Related work

The framework proposed in this thesis builds upon prior research in self-supervised learning, UAV-based object detection, and Vision-Language Models (VLMs). This section contextualizes the approach within these domains, highlighting the limitations of existing methods and how this thesis work advances the state of the art.

2.1 UAV Object Detection

Unmanned Aerial Vehicles (UAVs) have become increasingly valuable for applications such as environmental monitoring, disaster response, security surveillance, and traffic management. However, object detection from UAV poses distinct challenges arising from varying altitudes, dynamic camera angles, occlusions, and extreme weather conditions [14, 15]. These factors lead to substantial domain shifts, making it difficult for models trained on conventional object detection datasets to generalize effectively.

Aerial object detection is further challenged by environmental conditions such as variable lighting, fog, and snowfall, all of which reduce visibility and increase the likelihood of detection errors [2, 6, 16]. Additionally, objects captured by UAVs exhibit significant variations in scale and perspective, as changes in altitude directly affect their size and resolution [17]. For instance, vehicles observed from higher altitudes often occupy only a few pixels, making them difficult to distinguish from background elements. In addition, UAV cameras operate at different angles, causing objects to appear distorted

or rotated, further increasing the detection difficulty [18].

Various deep learning approaches have been proposed to tackle the challenges of UAV-based object detection. YOLO-based models are commonly employed due to their speed and efficiency, making them well-suited for real-time UAV applications [16, 19]. However, their performance deteriorates in adverse weather conditions, where objects often blend into the background [20]. Transformer-based models like DETR [21, 22] offer improved feature representation and exhibit greater robustness to occlusion and complex scenes. Yet, their high computational demands limit their applicability on UAVs with limited processing power. Meanwhile, Unsupervised Domain Adaptation (UDA) techniques [23] aim to bridge domain gaps and improve model generalization across environments, but typically involve fine-tuning or additional adaptation steps, which may not be feasible in UAV deployments.

2.2 Unsupervised and Self-Supervised Labeling

Unsupervised and self-supervised learning has witnessed widespread interest in its usage for reducing dependency on manually annotated datasets. Unsupervised labeling methods generally help by discovering structure in un-labelled data through clustering or generative modeling without any human-provided input. Classical methods include DeepCluster [24] and K-means-based feature learning. While fairly powerful, unsupervised techniques often lacks semantic grounding, limiting their usefulness in tasks requiring high-level understanding.

Self-supervised labeling, on the other hand, utilizes intrinsic properties of data such as spatial context, color channels, or temporal continuity to create pseudo-labels. This allows models to learn meaningful representations from unlabeled data with minimal human supervision. In computer vision, contrastive approaches like SimCLR [25], MoCo [26], and masked image modeling (e.g., MAE [27]) have achieved state-of-the-art results on downstream tasks.

In vision-language domains, models such as CLIP [28], ALIGN [29], and BLIP [30] learn representations from image-text pairs, enabling zero-shot pseudo-labeling by

interpreting textual prompts as queries for visual regions. Grounding DINO [31] enhances this ability by aligning open-vocabulary detection with DETR-based architectures. These Vision-Language Models (VLMs) serve as labelers in self-supervised pipelines, and have proven to be highly useful in applications where annotations are scarce or expensive.

2.3 Zero-shot object detection

Zero-shot object detection (ZSOD) aims to detect and localize objects in images without having seen labeled examples of those specific object classes during training. Unlike traditional supervised detectors that rely on a fixed set of predefined categories, zero-shot detectors generalize to novel classes by leveraging external knowledge, such as semantic embeddings or natural language descriptions. Early approaches to ZSOD relied on word embeddings and class similarity measures to transfer knowledge from seen to unseen categories. More recent methods, such as Grounding DINO [31] and GLIP [32], combine vision-language pretraining with object detection to enable open-vocabulary recognition. These models align image regions with textual queries, allowing detection of arbitrary objects described in natural language. While powerful, zero-shot detectors often produce noisy outputs and can miss objects with subtle visual features or ambiguous descriptions, motivating the need for additional filtering and refinement steps such as those proposed in this thesis.

2.4 Vision-Language Models as Judges

Recent advancements in Vision-Language Models (VLMs) have enabled multi-modal reasoning by integrating visual understanding with natural language processing. Models like PaliGemma2 [17] and Gemma3 [33] incorporate language based logic into visual tasks, allowing them to verify the presence of objects based on textual prompts. This capability positions them as effective semantic filters or “judges” for verifying the correctness of pseudo-labels in object detection pipelines.

In the context of pseudo-labeling, VLMs have been applied to open-world detection and class-incremental learning [34,35], where they assist in validating or refining bounding

box annotations. Instead of relying solely on detection confidence scores, VLMs reason over textual prompts to assess whether a detected object semantically matches a target category. This provides an additional verification layer that helps reduce false positives and enhance the precision of automatically generated labels.

Such verification is particularly valuable in complex settings like aerial imagery from Nordic regions, where snow, occlusions, or low visibility frequently lead to misclassifications. By incorporating VLM-based semantic filtering, the proposed framework ensures that only contextually relevant and semantically accurate detections are retained, improving both precision and robustness.

Moreover, VLMs are capable of interpreting objects under varying conditions, such as unusual angles, partial occlusions, or challenging lighting, which traditional detectors often misinterpret. This broader reasoning capacity enhances label quality in diverse real-world environments, especially in UAV-based applications.

2.5 Segmentation - spatial refinement/temporal smoothing

Semantic and instance segmentation have long been central tasks in computer vision. Traditional methods were based on fully supervised learning with dense, pixel-level annotations. Modern architectures like Mask R-CNN [36] have achieved high accuracy for segmentation, but still depends on large-scale labeled datasets. To overcome this, recent research has explored weakly-supervised and self-supervised segmentation methods using bounding boxes, or pseudo-labels generated via vision-language models.

Vision-Language Models (VLMs), such as CLIP [28] and Grounding DINO [31], have been adapted to generate segmentation masks from natural language prompts in a zero-shot manner. Additionally, models like SAM [37] provide universal segmentation capabilities by treating segmentation as a promptable task. In combination, VLMs and universal segmenters allow open-vocabulary segmentation pipelines where objects can be detected and segmented without the need for class labels.

SAM vs. Grounded-SAM

The Segment Anything Model (SAM) [37] introduced a promptable segmentation framework that can generate object masks from user-provided inputs such as points and boxes. Trained on over 1 billion masks, SAM generalizes well to a variety of objects and scenes in a zero-shot manner. However, SAM itself lacks semantic understanding; it does not interpret the meaning of text prompts or associate masks with object categories.

Grounded-SAM [38] addresses this limitation by integrating SAM with the Vision-Language Model Grounding DINO [31]. This combination allows the system to ground natural language phrases (e.g., "a red car" or "person on the left") into specific image regions and then extract precise segmentation masks using SAM. As a result, Grounded-SAM supports open-vocabulary segmentation, enabling segmentation based on rich textual queries without predefined class labels.

While SAM excels in producing high-quality, edge-aligned masks, its effectiveness depends on well-placed prompts. Grounded-SAM, on the other hand, automates the prompt generation using vision-language grounding, which improves usability and supports scalable pseudo-label generation. This makes Grounded-SAM particularly valuable for self-supervised or unsupervised labeling pipelines where semantic grounding and mask quality are both critical.

From SAM to SAM2 and Grounded-SAM2

While the original Segment Anything Model (SAM) [37] introduced a promptable framework for high-quality image segmentation, it operated solely on static images and lacked temporal modeling capabilities. SAM2 [39] extends this framework to video, enabling promptable segmentation across frames with improved spatial and temporal coherence. SAM2 integrates temporal attention mechanisms to propagate object masks reliably across time, making it more suitable for applications in video-based labeling and tracking.

To incorporate semantic understanding, Grounded-SAM2 combines SAM2 with the vision-language model Grounding DINO [31]. In this setup, the vision-language model grounds open-vocabulary text queries (e.g., "the red bicycle" or "a person wearing

glasses") to spatial regions in the video frames. These regions serve as prompts for SAM2, which then propagates and refines the masks across the video sequence. This allows for open-vocabulary, temporally consistent segmentation without requiring predefined categories or dense annotations.

Compared to Grounded-SAM, which uses the original SAM for single-image segmentation, Grounded-SAM2 introduces significant advantages such as improved mask continuity, temporal stability, and reduced need for repeated prompting. This makes Grounded-SAM2 especially valuable in self-supervised or unsupervised pseudo-labeling pipelines where video consistency is essential.

Summary. These recent developments in universal and open-vocabulary segmentation demonstrate the growing role of modular segmentation models like SAM2 in both spatial refinement and temporal smoothing. Building on these advancements, this thesis explores how such models can be effectively integrated with vision-language guidance for generating high-quality pseudo-labels in unsupervised settings.

Chapter 3

Methodology

This chapter presents the framework developed in this thesis. It begins with a high-level overview of the system architecture and then proceeds to describe each component in detail.

3.1 Solution overview

An overview of the complete system is illustrated in Figure 3.1. The diagram outlines all layers of the framework, the state-of-the-art models employed in each layer, and their respective high-level goals.

The first layer of the proposed framework serves as the base object detector. It takes an image and a textual prompt as input and produces bounding boxes corresponding to objects in the image that confirms to the given prompt. This layer employs Grounding DINO [31] as the underlying model to perform open-vocabulary object detection based on the input query.

The second layer of the framework performs semantic filtration of the detected objects. It takes the cropped image regions corresponding to the bounding boxes generated in Layer 1 and evaluates their alignment with the original prompt. This is achieved using a Vision-Language Model (VLM) to measure semantic similarity between the prompt and each object crop. In this work, VLMs such as Gemma3 [33] and PaliGemma2 [17] were evaluated for this purpose. While the overall structure of the layer remains the

same, the choice of VLM affects the precision and recall of the filtration process. Results comparing the two models are discussed in Section 4: *Results and Discussions*.

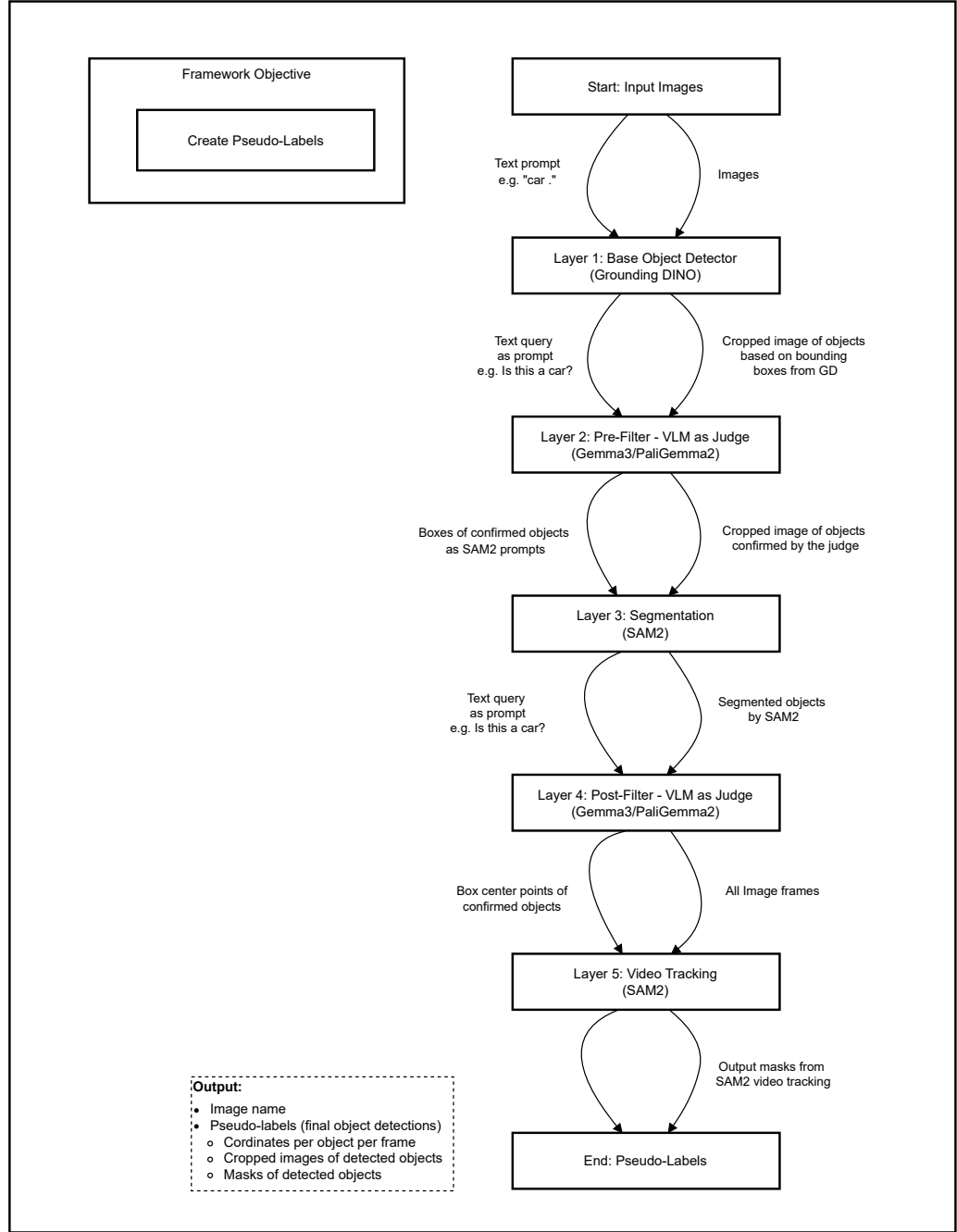


Figure 3.1: Framework overview

The third layer of the framework applies image level segmentation using SAM2. It receives the filtered bounding boxes from the previous layer and uses them as prompts

to segment the corresponding object regions, refining spatial boundaries and producing precise object masks.

The fourth layer performs post-filtering on the segmented objects to further verify semantic relevance. Each mask obtained from SAM2 is cropped from the image and re-evaluated using a Vision-Language Model. Similar to layer 2 (i.e. Pre-Filter layer), both Gemma3 and PaliGemma2 were tested in this stage to assess alignment between the segmented region and the original prompt. This step serves as a final validation to eliminate false positives and ensure that only semantically accurate masks are retained.

The fifth layer enhances temporal consistency using SAM2 based video tracking. After high-quality detections are obtained from the fourth layer, SAM2 propagates these across neighboring frames to recover missed instances. By tracking objects over time, the layer improves recall and ensures continuity, producing temporally coherent pseudo-labels.

3.2 Detailed Description of the Framework

A detailed view of the proposed framework is illustrated in Figure 3.2. The diagram outlines the flow of data across all layers, highlighting the role of each component, the models used, and how outputs from one layer serve as inputs to the next.

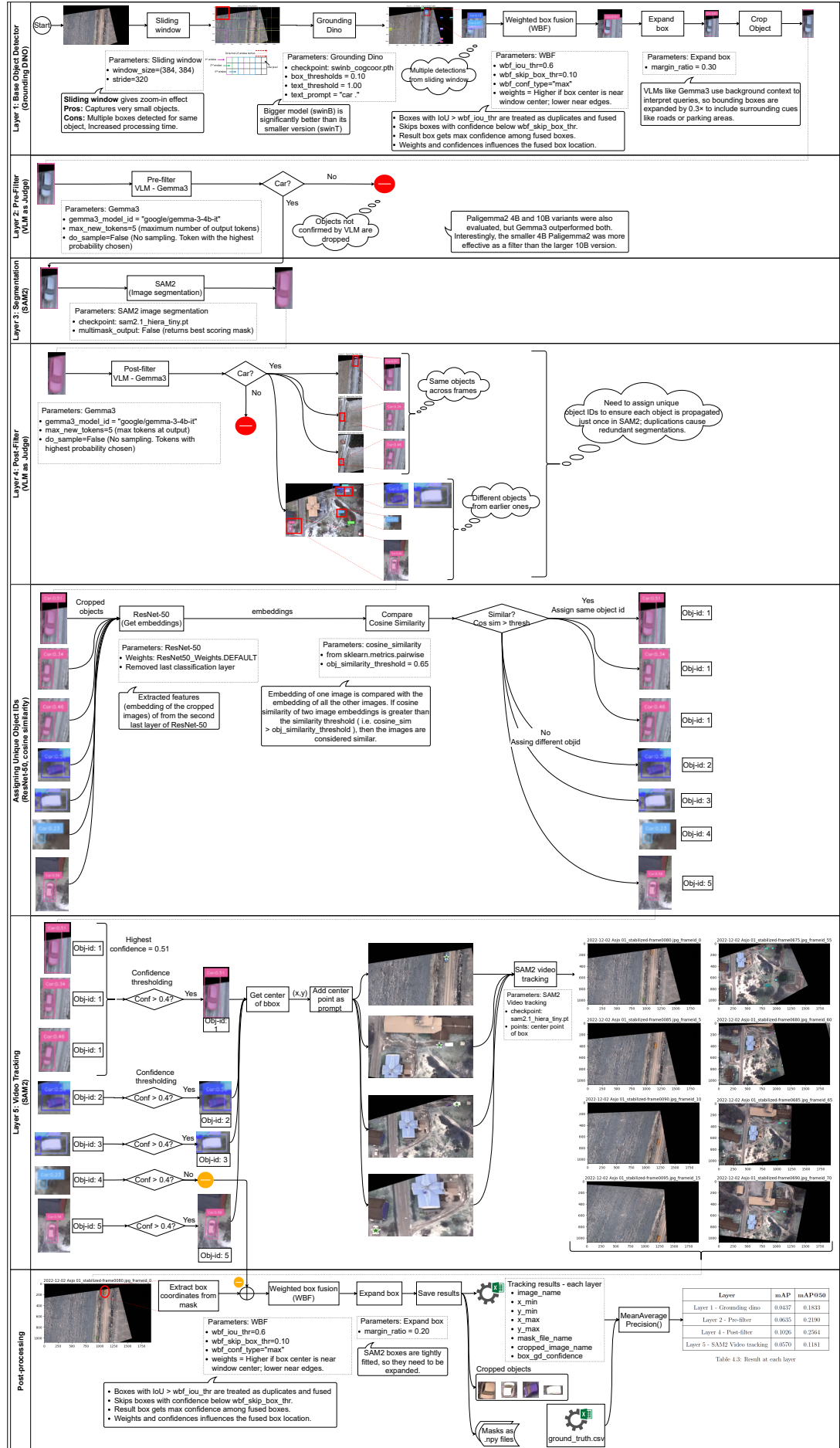


Figure 3.2: Framework detail

Subsequent sections describe each layer in detail, including its function and design choices implemented.

3.2.1 Layer 1: Base Object Detector

Purpose: As shown in Figure 3.3, this layer serves as the base object detector and plays a critical role in the framework. Its objective is to identify all relevant objects in the image that correspond to the input textual prompt. Since only objects detected at this stage are passed to subsequent layers, any missed detections by this layer, cannot be recovered by later layers. Therefore, in applications where recall is a priority, using a lower bounding box threshold at the base detection layer is recommended to maximize detection coverage.

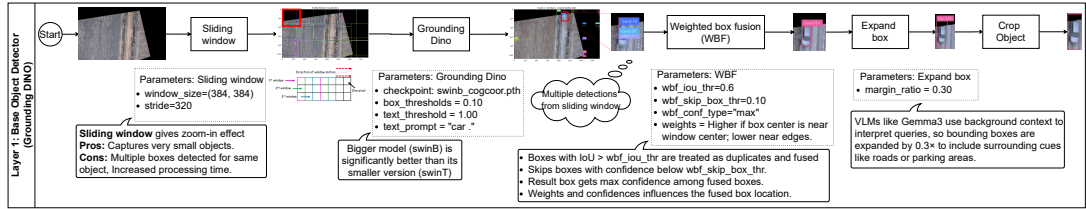


Figure 3.3: Layer 1 - Base object detection with Grounding DINO

Input: The input to this layer is a raw image. The Grounding DINO model is used for open-vocabulary object detection and takes additional parameters such as the text prompt, box threshold, and text threshold.

Process: Each image is processed using a sliding window approach to create multiple overlapping image crops, allowing Grounding DINO to detect small objects more effectively. Each windowed crop is passed to the model, and the resulting bounding box coordinates are adjusted relative to the original image dimensions. Since this method can result in multiple overlapping detections of the same object, the Weighted Boxes Fusion (WBF) algorithm is applied to remove duplicates. The remaining unique boxes after de-duplication, are then expanded by a fixed factor to preserve context (e.g., road, background) around the object, which is beneficial for downstream tasks. More details about the parameters used can be found in Figure 3.3.

Output: A set of expanded, non-overlapping bounding boxes representing the final

object detections from layer 1.

Remarks: The larger Grounding DINO variant (Swin Transformer Base, SwinB) has been observed to perform better than the smaller variant (Swin Transformer Tiny, SwinT). Additionally, WBF is preferred over Non-Maximum Suppression (NMS) due to its superior handling of overlapping detections.

3.2.2 Layer 2: Pre-Filter - VLM as Judge

Purpose: As illustrated in Figure 3.4, the purpose of this layer is to filter out detections that do not correspond to the intended target object. In the previous layer, a low box threshold was used in Grounding DINO to maximize recall, which also introduced significant noise. This layer aims to reduce false positives and suppress noisy detections before proceeding to the next layer.

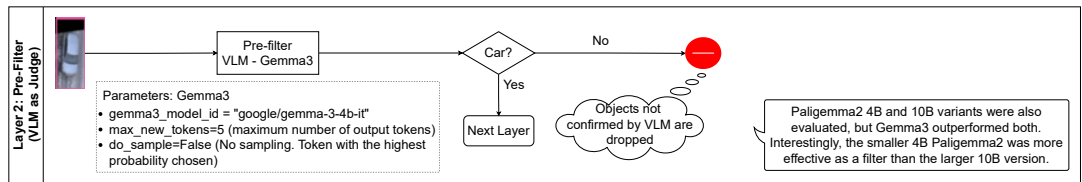


Figure 3.4: Layer 2 - Pre-Filter - VLM as Judge

Input: Cropped images that are based on the bounding boxes obtained from Layer 1. These cropped images are passed to a Vision-Language Model (VLM) along with a system prompt and user prompt. For example, the system prompt used was "You are a vision-language expert. Identify cars in aerial drone images.", and the user prompt was "This is a image crop of a drone-captured aerial image. Does it contain the top view of a car? Do not consider buildings, houses, shadows or any other objects. Only consider car. Answer strictly with either 'yes' or 'no' in lowercase."

Process: The instruction-tuned Gemma3-4B model is used in this layer to perform semantic verification. It receives the cropped image region along with a system prompt and a user prompt, enabling the model to reason about whether the visual content aligns with the intended object description. This prompt based interaction helps filter out semantically irrelevant detections.

Output: The model outputs a binary response "yes", if the image region matches the

prompt, and “no” otherwise. Only detections receiving a “yes” are passed to the next layer.

Remarks: In addition, VLM PaliGemma2 was also tested and is discussed in Section 4: *Results and Discussions*.

3.2.3 Layer 3: Segmentation

Purpose: This layer, as shown in Figure 3.5 performs segmentation on the filtered object crops to extract precise object masks. These segmented objects are further refined in the next layer through an additional round of semantic filtration, aimed at further reducing false positives.

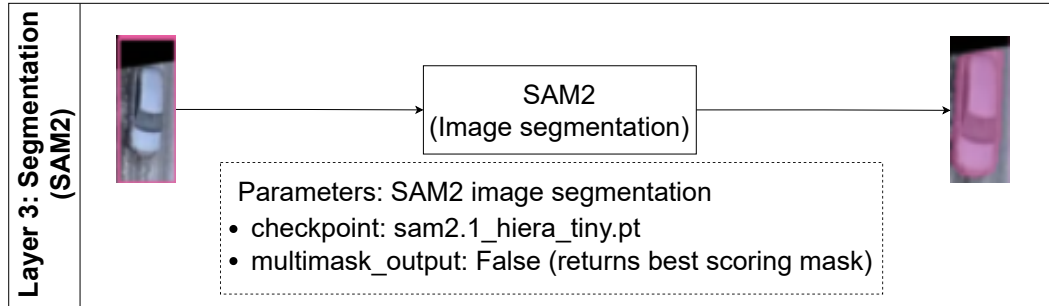


Figure 3.5: Layer 3: Segmentation with SAM2

Input: Cropped images of objects that were confirmed as valid detections by the previous filtration layer.

Process: Each cropped image, along with its corresponding bounding box, is passed to a SAM2 based image segmentation module. SAM2 generates high-resolution segmentation masks that tightly fit the object within each crop.

Output: A set of segmentation masks corresponding to the filtered objects.

Remarks: The first filtration was applied to bounding boxes generated by Grounding DINO. This second round of filtration (in the next layer) operates on the segmented object masks created in this layer, allowing for more accurate semantic verification and reducing noise in the pseudo-labels.

3.2.4 Layer 4: Post-Filter - VLM as Judge

Purpose: Similar to Layer 2 Pre-Filter, the objective of this layer shown by Figure 3.6 is to eliminate irrelevant or incorrectly detected objects. However, this filtration is performed on segmented object masks rather than bounding boxes, allowing for more precise semantic verification.

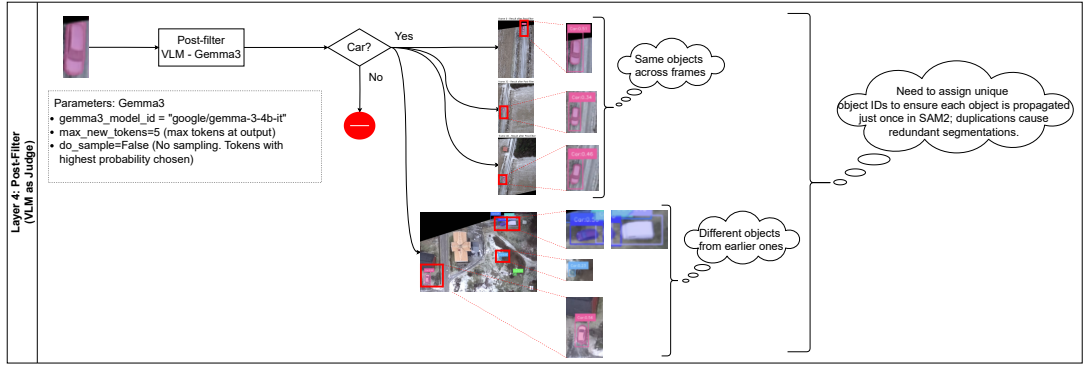


Figure 3.6: Layer 4: Post-Filter - VLM as Judge

Input: Segmented object masks generated in Layer 3. The Vision-Language Model (VLM) configuration used here is identical to that used in Layer 2 (i.e., the Pre-Filter layer).

Process: Each segmented object is passed to the Gemma3 Vision-Language Model, along with the associated prompts. This step evaluates whether the segmented region semantically aligns with the target object described in the prompt.

Output: A binary response: "yes" if the segmented object matches the target description, and "no" otherwise. Only detections receiving a “yes” are passed to the next layer.

Remarks: The VLM model configuration remains the same as in Layer 2 (i.e., the Pre-Filter layer); however, the input here is a segmented object mask instead of a bounding box crop.

3.2.5 Assigning Unique Object IDs

Purpose: The primary objective of this layer as illustrated in Figure 3.7 is to uniquely identify objects across frames and assign them with object ids.

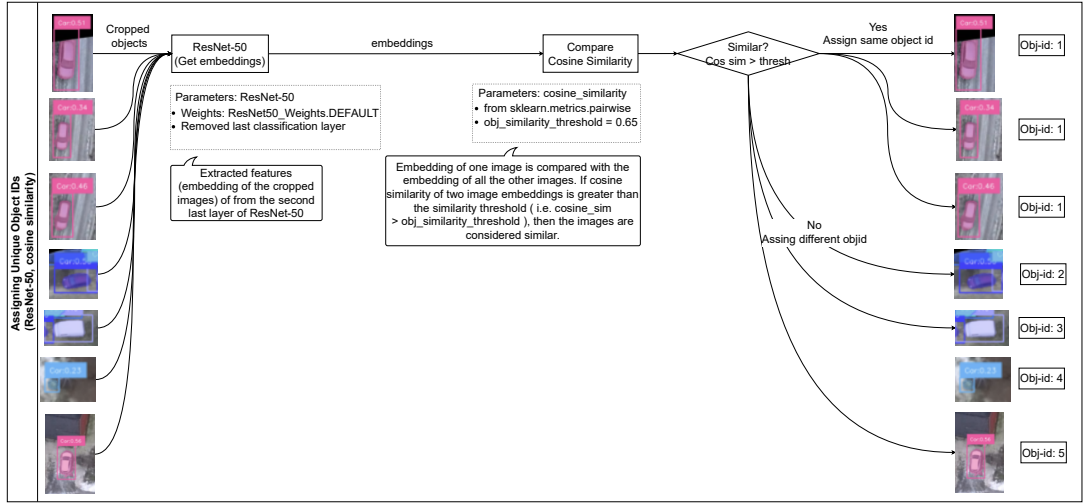


Figure 3.7: Assigning Unique Object IDs

Input: Segmented object masks from the previous layer that were positively confirmed as target objects.

Process: This layer uses ResNet-50 model to extract feature embeddings for each detected object. The final classification layer is removed from ResNet-50 model, so that the embeddings can be taken from the second last layer. Cosine similarity is then used to compare these embeddings, and objects with a similarity score above 0.65 are treated as the same instance and assigned same object ID.

Output: Unique object ids for each detected objects. Similar objects get same object id.

Remarks: If this layer is skipped, each occurrence of an object may be propagated multiple times through SAM2 video tracking, leading to redundant and duplicate masks. Since video propagation is computationally intensive, avoiding such redundancy is essential for efficient execution.

3.2.6 Layer 5: Video Tracking

Purpose: Figure 3.8 illustrates the fifth layer of the pipeline. Its primary objective is to improve recall by recovering missed detections across video frames through temporal mask propagation.

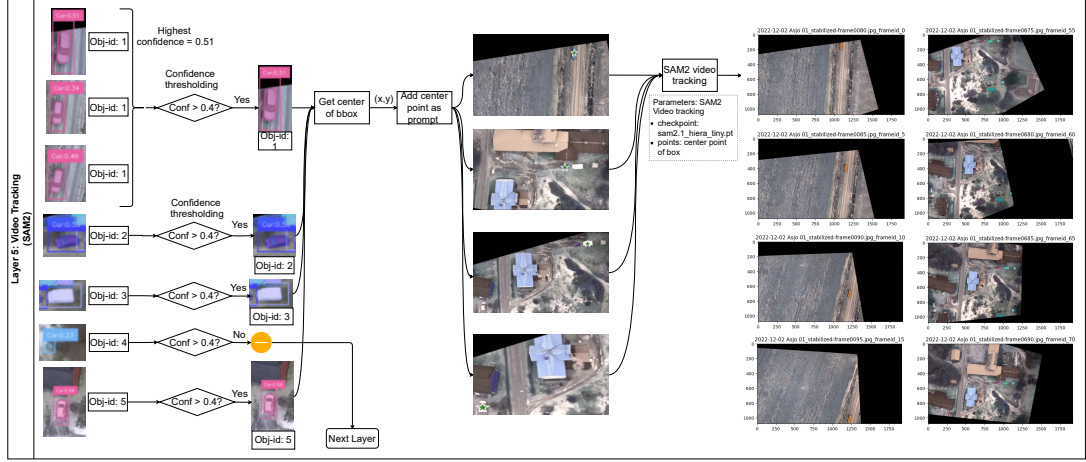


Figure 3.8: Layer 5: Video Tracking using SAM2

Input: Bounding boxes of unique positive detections from the previous layer.

Process: For each object ID, only a single instance is selected to initiate propagation through the SAM2 video tracking module. When multiple detections share the same object ID, the one with the highest confidence score is chosen. For example, as shown in Figure 3.8, for object ID 1, the detection with the highest confidence score of 0.51 was selected for propagation. This helps prevent duplicate propagation and reduces unnecessary computational overhead.

To avoid propagating noisy detections, confidence thresholding is applied at this stage. Only high-confidence detections are passed to SAM2 for video tracking. For example, as shown in Figure 3.8, only detections with a confidence score greater than 0.4 were selected for propagation. Detections falling below the threshold are excluded from propagation but are retained for further processing in the next post-processing layer, where they can still contribute to the final results.

For the selected high-confidence detections, the center points of their corresponding bounding boxes are computed and used as prompts for SAM2 propagation. These prompts are then propagated across all frames, allowing the model to recover missed detections across frames.

Output: Segmentation masks across video frames, including masks for objects that were missed in some frames by earlier layers.

Remarks: While SAM2 video tracking enhances recall, it may also propagate noise if

earlier layers do not effectively filter out false positives. For applications where occasional false positives are acceptable, this trade-off is less critical. However, in precision-sensitive use cases, strong filtering in earlier stages is recommended before enabling this layer.

3.2.7 Post Processing

Purpose: The purpose of this layer, as illustrated in Figure 3.9, is to refine and finalize the detection results.

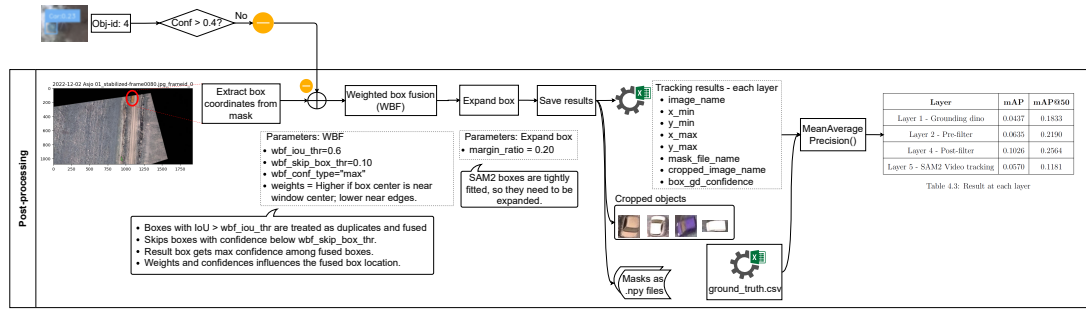


Figure 3.9: Post Processing

Input: Propagated detections from SAM2 and un-propagated low-confidence boxes from the previous layer.

Process: The post-processing layer begins by extracting bounding boxes from the segmentation masks produced by the SAM2 video tracking module. These boxes are then combined with the low-confidence detections that were previously excluded from propagation. The merged set of boxes is passed through Weighted Box Fusion (WBF) to remove duplicates and consolidate overlapping detections.

Since the bounding boxes derived from SAM2 masks tend to be tightly fitted around objects, a box expansion step is applied to better match standard box sizes, preventing unfair penalization in mAP evaluation despite correct detections.

At this stage, various outputs are also saved for analysis, including cropped object images, segmentation masks, and intermediate results from each layer of the pipeline. In addition, evaluation metrics such as mean Average Precision (mAP) and mAP@50 are computed for each layer to assess performance throughout the pipeline.

Output: Results from the pipeline including final bounding boxes, masks, cropped object images, and per-layer evaluation metrics (mAP and mAP@50).

Remarks: Outputs from this layer are logged per frame for reproducibility and detailed analysis.

Chapter 4

Results and Discussions

4.1 Metrics

This work implements a framework for generating pseudo-labels from unannotated image data. To evaluate its effectiveness, the mean Average Precision (mAP) metric is used which is calculated as the mean of Average Precision values over multiple Intersection over Union (IoU) thresholds. Since this thesis focuses on a single object class "car", the evaluation focuses solely on that category. To assess the performance of the framework, both the overall mAP and mAP@50 metrics, as described in Table 4.1, were recorded during test runs.

Table 4.1: Metrics used for evaluation

Metric	Description
mAP	Average of Average Precision values at IoU thresholds ranging from 0.50 to 0.95 in steps of 0.05
mAP@50	Average Precision calculated at $\text{IoU} \geq 0.50$

4.2 Dataset

The proposed solution was tested on a set of images belonging to Nordic vehicle dataset (NVD) developed by Mokayed et al. [6]. Nordic vehicle dataset (NVD) comprises of videos of aerial footage captured in the north of Sweden, with mostly snowy weather

conditions. NVD dataset provides high resolution video/images captured by UAV in real Scandinavian winter encompassing low light and sub-zero conditions. Recorded at flight height varying between 120 and 250 meters, includes video resolution varying between 1920 x 1080 and 3840 x 2160. Unlike VisDrone [40] and UAVDT [41], NVD has special focus on vehicles anywhere in the picture and not just on roads, considers varying snow cover on vehicles as well as background. NVD focuses on the extreme weather conditions which is lacking in other publicly available datasets. By providing challenging weather specific data, NVD enables for research and drives solutions for actual industrial adoption in geographies similar to north of scanidnavia where today's state of the art detection models struggle to perform reliably.

4.3 Evaluation

To systematically evaluate the performance of the proposed unsupervised annotation pipeline, test sets of increasing size and complexity were used. The initial evaluation, referred to as the 100-Image test, was conducted on a set of 100 images containing small-sized cars to validate baseline functionality. To further assess scalability and robustness, the evaluation was extended to a larger and more diverse dataset of 500 images, referred to as the 500-Image test. Each test is discussed in detail in the subsections below, along with the corresponding quantitative results and key observations.

Base Configuration for Evaluation

The following configuration was used as the base setup for evaluating the framework across all test runs. This configuration applies to both the 100-Image and 500-Image test sets. Where deviations from this setup were made for specific experiments (e.g., confidence thresholding or prompt variations), they are explicitly mentioned in the corresponding sections.

Table 4.2: Base Configuration for Evaluation

Parameter	Value	Description
gd_checkpoint	SwinB	Grounding DINO checkpoint
gd_box_thresholds	0.10	Grounding DINO - box threshold
gd_text_threshold	1.00	Grounding DINO text threshold
gd_text_prompt	"car ."	Grounding DINO text prompt
window_size	(384, 384)	Window size for sliding window
stride	320	Stride for sliding window
de_dup_boxes_method	"wbf"	Weighted box fusion for deduplication of detected objects from sliding Grounding DINO
vlm_model	"gemma3"	When Gemma3 is used as VLM for filtration
gemma3_model_id	"google/gemma-3-4b-it"	Gemma3 model
g3_prompt_prefilter	This is a image crop of a drone-captured aerial image. Does it contain the top view of a car? Do not consider buildings, houses, shadows or any other objects. Only consider car. Answer strictly with either 'yes' or 'no' in lowercase.	Prompt for Pre-Filter when Gemma3 is used as VLM for filtration
g3_prompt_postfilter	Same as g3_prompt_prefilter	Prompt for Post-Filter when Gemma3 is used as VLM for filtration

Parameter	Value	Description
g3_sys_content_prompt	You are a vision-language expert. Identify cars in aerial drone images.	System Prompt for Gemma3
vlm_model	"PaliGemma2"	When PaliGemma2 is used as VLM for filtration
pg2_model_id	"google/paligemma2-3b-pt-224"	PaliGemma2 model
pg2_prompt_prefilter	"<image> The object in the image is not a car. Answer with 'yes' or 'no'."	Prompt for Pre-Filter when PaliGemma2 is used as VLM for filtration
pg2_prompt_postfilter	Same as pg2_prompt_prefilter	Prompt for Post-Filter when PaliGemma2 is used as VLM for filtration
sam2_checkpoint	sam2.1_hiera_tiny	SAM 2 checkpoint
resnet50_model	ResNet50_Weights.DEFAULT	Model to extract embeddings of detected objects
obj_similarity_threshold	0.65	Objects having cosine similarity greater or equal to obj_similarity_threshold are considered similar

4.3.1 Initial Evaluation on 100 Images

The proposed framework was evaluated on a curated test set of 100 images from the Nordic Vehicle Dataset (NVD), focusing specifically on scenarios involving small cars. These instances are particularly challenging for detection due to their limited pixel footprint and background complexity.

Initial evaluations revealed that the framework struggled to detect small objects effectively. Even after lowering the base detector’s box threshold (Grounding DINO)

to 0.10, performance remained suboptimal. To address this issue, a sliding window approach was introduced before passing the image to Grounding DINO. This technique created a zoom-in effect, which significantly improved the model’s ability to detect small vehicles.

Two windowing strategies were considered: non-overlapping and overlapping. Non-overlapping windows ensure no redundancy between image crops and are more computationally efficient; however, they risk missing objects that lie on the edges of window boundaries. On the other hand, overlapping windows increase redundancy and computational load but offer better coverage, making them more reliable for detecting small or partially visible objects.

Given the goal of maximizing recall for small cars, the overlapping strategy was chosen. While it led to duplicate detections due to object repetition across window regions, this was considered preferable to the risk of entirely missing detections. To resolve the issue of duplicates, the Weighted Boxes Fusion (WBF) technique was implemented, as described in earlier sections.

The subsections below present detailed results from the 100-Image test set focused on detecting very small vehicles.

Test Set – 100-Image - Result

Table 4.3 presents the detection performance at each layer of the proposed pipeline, evaluated on a challenging subset of 100 images containing small cars and the configuration described in Table 4.2. The Vision-Language Model used in Layer 2 (Pre-Filter) and Layer 4 (Post-Filter) was Gemma3.

Table 4.3: Result at each layer - 100-Image

Layer	mAP	mAP@50
Layer 1 - Base Object Detector	0.0437	0.1833
Layer 2 - Pre-Filter	0.0635	0.2190
Layer 4 - Post-Filter	0.1026	0.2564
Layer 5 - Video Tracking	0.0570	0.1181

The results show a clear improvement in both mAP and mAP@50 as the pipeline progresses through the filtering and refinement stages.

Layer 1 Base Object Detector, provides the initial detections using zero shot open-vocabulary detection capability of Grounding DINO. While it serves as the foundation, the performance is modest (mAP = 0.0437), reflecting the difficulty of detecting small objects in complex aerial scenes.

Layer 2 Pre-Filter, introduces semantic filtering using vision-language model Gemma3, which helps eliminate irrelevant detections and enhances precision. This results in a noticeable boost in both mAP and mAP@50.

Layer 4 Post-Filter, which filters based on segmented objects instead of bounding boxes, shows the highest detection performance (mAP = 0.1026 and mAP@50 = 0.2564). This indicates that segmentation based filtering is more effective for precise object identification, especially for small-scale targets.

Layer 5 Video Tracking, shows a drop in performance, with both mAP and mAP@50 decreasing. While video propagation helps recover missed detections by spreading object masks across frames, it can also amplify existing noise from earlier layers. Noisy or incorrect detections introduced earlier in the pipeline may get propagated temporally, leading to a reduction in overall precision. This highlights the need for more selective propagation strategies to balance recall gains without degrading output quality.

Overall, the table validates the effectiveness of multi-stage semantic filtering, with the highest gains observed when segmentation masks are used for verification.

Improving SAM2 Video Tracking with Confidence thresholding

To improve the performance of Layer 5 Video tracking, an additional refinement strategy of propagating only high-confidence detections was explored. The motivation behind this approach was to reduce the spread of noisy or low-confidence predictions across video frames, which could otherwise degrade the precision of the results.

Experiments were conducted using two propagation confidence thresholds (i.e., `prop_th` = 0.4, 0.5), alongside the baseline setting with no confidence thresholding (`prop_th` = NA). As shown in Table 4.4, selectively propagating only those detections

with confidence scores greater than 0.4 and 0.5 resulted in significant performance improvements in Layer 5.

Most notably, setting the threshold to 0.5 yielded the highest performance, achieving an mAP of 0.0920 and mAP@50 of 0.2341. This is a substantial improvement compared to the baseline (no threshold), where mAP was only 0.0570 and mAP@50 was 0.1181. The results clearly indicate that limiting propagation to high-confidence detections reduces temporal noise and enhances the quality.

Table 4.4: Effect of confidence thresholding during SAM2 Video Tracking (Layer 5). Increasing threshold improves mAP@50 by suppressing noisy propagation.

Layer	prop_th = NA		prop_th = 0.4		prop_th = 0.5	
	mAP	mAP@50	mAP	mAP@50	mAP	mAP@50
Layer 1 - Base Object Detector	0.0437	0.1833	0.0437	0.1833	0.0437	0.1833
Layer 2 - Pre-Filter	0.0635	0.2190	0.0635	0.2190	0.0635	0.2190
Layer 4 - Post-Filter	0.1026	0.2564	0.1026	0.2564	0.1026	0.2564
Layer 5 - Video Tracking	0.0570	0.1181	0.0829	0.1834	0.0920	0.2341

Motivation for Using Sliding Window in Base Detection

Initial experiments using the full image as input to Grounding DINO resulted in extremely poor performance with a mAP of just 0.0052 and mAP@50 of 0.0099. This was primarily because small objects were not sufficiently visible or distinguishable when the entire image was processed at once.

To overcome this limitation, a sliding window approach was adopted, dividing each image into overlapping windows of size (384, 384) with a stride of 320. This significantly improved detection performance, raising the mAP to 0.0920 and mAP@50 to 0.2341, as shown in Table 4.5. The results clearly indicate that processing smaller image regions helps the model detect fine details and small sized objects that would otherwise be missed in a full-image pass.

Table 4.5: Effect of Sliding Window on Detection Performance. Sliding window significantly improves detection of small objects (mAP@50 from 0.0099 $\rightarrow$ 0.2341).

Layer	No Sliding Window		Sliding Window	
	mAP	mAP@50	mAP	mAP@50
Final layer	0.0052	0.0099	0.0920	0.2341

Grounding DINO Box Threshold Comparison and Its Implications

Table 4.6 presents the detection performance, measured in terms of mAP and mAP@50, across different layers of the proposed framework and for three values of the Grounding DINO box threshold: 0.10, 0.15, and 0.20. The evaluation was conducted on a challenging set of 100 images from NVD, containing small cars.

This result clearly shows that the choice of box threshold in the base detector has a significant impact on overall performance. With lower box threshold of 0.10 for Grounding DINO at layer 1, mAP and mAP@50 scores has been consistently higher across all layers. This suggests that a more permissive setting in the base detector helps retain a greater number of true positives, even if it introduces some noise. In contrast, higher thresholds (e.g., 0.20) lead to a drop in performance, as stricter initial filtering causes many valid detections especially small objects to be discarded early. Since missed detections at this stage cannot be recovered by downstream layers, keeping a lower threshold in the base detector provides a stronger foundation for subsequent refinement and overall detection coverage.

Table 4.6: Effect of varying Grounding DINO box thresholds on detection performance. Lower box threshold (0.10) achieves highest overall mAP and mAP@50 across layers.

Layer	box_th = 0.10		box_th = 0.15		box_th = 0.20	
	mAP	mAP@50	mAP	mAP@50	mAP	mAP@50
Layer 1 - Base Object Detector	0.0437	0.1833	0.0434	0.1800	0.0396	0.1617
Layer 2 - Pre-Filter	0.0635	0.2190	0.0616	0.2151	0.0410	0.1581
Layer 4 - Post-Filter	0.1026	0.2564	0.1013	0.2529	0.0252	0.0536
Layer 5 - Video Tracking	0.0829	0.1834	0.0761	0.1709	0.0729	0.1633

Effect of Box Threshold on Pipeline Processing Time

While lower box thresholds improve detection performance by preserving more true positives, they also lead to an increase in the number of candidate boxes, including noisy or irrelevant detections. As a result, downstream layers must process a larger volume of data, which increases the overall computational cost and processing time of the pipeline as given in table 4.7. This trade-off highlights the need to balance recall and efficiency when selecting an appropriate detection threshold, particularly in resource-constrained or real-time deployment scenarios.

Table 4.7: Effect of lowering Grounding DINO box threshold on overall pipeline processing time. Lower thresholds increase processing cost.

Box Threshold	Pipeline Processing Time (mins:secs)
0.10	28:41
0.15	13:07
0.20	08:46

Grounding DINO Text Threshold Comparison and Its Implications

To evaluate the influence of the text threshold parameter in Grounding DINO (GD), experiments were conducted using three different values of text threshold: 0.00, 0.50, and 1.00 and with text prompt of "car.". The results, summarized in Table 4.8, indicate that varying the text threshold has no impact on base object detection performance at layer 1.

Table 4.8: Effect of Grounding DINO text threshold on base object detection performance — no measurable impact observed across thresholds.

Layer	text_th = 0.00		text_th = 0.50		text_th = 1.00	
	mAP	mAP@50	mAP	mAP@50	mAP	mAP@50
Layer 1 - Base Object Detector	0.0437	0.1833	0.0437	0.1833	0.0437	0.1833

De-duplication of Boxes: WBF vs. NMS Comparison

An important comparison in the pipeline involved evaluating the effectiveness of Weighted Box Fusion (WBF) versus Non-Maximum Suppression (NMS) for bounding box de-

duplication. Since each frame was divided into overlapping windows (384×384 with a stride of 320) before being processed by Grounding DINO (GD), many objects appeared in multiple windows, resulting in duplicate detections.

To address this, both WBF and NMS were applied immediately after GD at Layer 1. The rest of the pipeline remained unchanged. The test was run twice, once with Gemma3 and once with PaliGemma2 as the vision-language model at filtration layers.

As shown in Table 4.9, which presents the final output of the pipeline, WBF outperformed NMS in terms of mAP and mAP@50. This trend was observed across both pipeline configurations: one using Gemma3 and the other using PaliGemma2 as the vision-language model, indicating that WBF is more effective at consolidating overlapping detections in the sliding window setup.

Table 4.9: Effect of de-duplication method: WBF outperforms NMS for both Gemma3 and PaliGemma2.

Test description	Gemma3		PaliGemma2	
	mAP	mAP@50	mAP	mAP@50
WBF for de-duplication	0.0883	0.2010	0.0897	0.2213
NMS for de-duplication	0.0771	0.1684	0.0759	0.1634

Is a VLM needed as a judge?

One of the primary objectives of this thesis was to evaluate whether Vision-Language Models (VLMs) are essential as judges within the proposed multi-layer unsupervised annotation pipeline. Table 4.3 presents the performance metrics (mAP and mAP@50) at each processing layer when tested on 100-Image test set.

The results show that incorporating a VLM-based pre-filter at Layer 2 leads to improved performance over the baseline detector, Grounding DINO, used at Layer 1. Specifically, mAP increased from 0.0437 to 0.0635, and mAP@50 improved from 0.1833 to 0.2190. This suggests that the VLM filter effectively suppresses irrelevant detections while preserving meaningful ones.

Further gains are observed at Layer 4, where a post-filtering step using the VLM boosts mAP to 0.1026 and mAP@50 to 0.2564, representing more than double the original

mAP. This layer differs from Layer 2 in that filtering is performed on segmentation masks rather than bounding boxes. The higher performance indicates that segmentation based filtering allows the VLM to make more informed judgments about object relevance, as it has access to finer spatial detail and context compared to coarse bounding boxes.

These findings highlight two key insights: (1) VLMs significantly enhance detection performance when used as judges, and (2) segmentation based filtration is more effective than bounding box based filtration.

Therefore, this work concludes that VLMs are indeed necessary as judges in the proposed system, and their utility is maximized when paired with high-resolution segmentation inputs.

VLMs as judge - Gemma3 vs PaliGemma2 comparison

In this thesis, a comparative study was also conducted between two vision language models - Gemma3 and PaliGemma2. The goal was to find the model that provides better filtration capabilities. Results indicates that Gemma3 successfully detects many target objects but also includes significant noise resulting in high recall but low precision. For uses cases where it is more important to detect all relevant objects, even at the cost of false positives, Gemma3 is recommended. On the other hand, PaliGemma2 results in high precision by filtering out noise but also fails to detect many target objects. So for uses cases where precision is more important than recall, PaliGemma2 is a preferred choice. From table 4.10, it can be concluded that Gemma3 provides better filtration capabilities.

Table 4.10: Comparison of VLMs as judges: Gemma3 consistently outperforms PaliGemma2 in both pre-filter and post-filter stages.

Layer	Gemma3		PaliGemma2	
	mAP	mAP@50	mAP	mAP@50
Layer 1 - Base Object Detector	0.0437	0.1833	0.0437	0.1833
Layer 2 - Pre-Filter	0.0635	0.2190	0.0556	0.1935
Layer 4 - Post-Filter	0.1026	0.2564	0.0832	0.2071
Layer 5 - Video Tracking	0.0829	0.0.1834	0.0868	0.2117

There is also a noticeable difference in computation time. With identical configurations, the pipeline runs in 16 minutes 43 seconds using PaliGemma2, while it takes 30 minutes 21 seconds with Gemma3, making Gemma3 approximately 1.8 times slower than PaliGemma2.

Effect of de-duplicating final detections

Table 4.11 illustrates the effect of applying a final de-duplication step using Weighted Box Fusion (WBF) after Layer 5 Video Tracking. As observed, the performance improves consistently after this additional refinement.

When tested with Gemma3 as VLM, the mAP increases from 0.0829 to 0.0883. Similarly, for the pipeline with PaliGemma2 as VLM, mAP improves from 0.0868 to 0.0897.

These gains demonstrate that even after multiple layers of filtering and video propagation, redundant detections can persist. Incorporating WBF at the final stage helps in consolidating overlapping detections, reducing noise, and thereby improving overall pseudo-labeling quality.

Table 4.11: Effect of final de-duplication (WBF) after Layer 5 Video Tracking : improves overall mAP

Layer	Gemma3		PaliGemma2	
	mAP	mAP@50	mAP	mAP@50
Layer 1 - Base Object Detector	0.0437	0.1833	0.0437	0.1833
Layer 2 - Pre-Filter	0.0635	0.2190	0.0556	0.1935
Layer 4 - Post-Filter	0.1026	0.2564	0.0832	0.2071
Layer 5 - Video Tracking	0.0829	0.1834	0.0868	0.2117
Post Processing - De-duplicate (WBF)	0.0883	0.2010	0.0897	0.2213

4.3.2 Prompt Sensitivity

Prompt phrasing plays a critical role in determining the effectiveness of vision-language models (VLMs) and open-vocabulary object detectors. Differences in wording, polarity, and contextual details can significantly impact detection and filtering performance. To

systematically analyze this sensitivity, experiments were conducted across three models: Grounding DINO, PaliGemma2, and Gemma3.

This section presents definitions for key prompt attributes, Polarity and Type, and provides a comparative analysis of prompt sensitivity for each model, highlighting how carefully engineered prompts can guide model behavior and improve pseudo-labeling outcomes.

Prompt Polarity and Type Definitions

The following two tables define the Prompt Polarity and Prompt Type categories used across all prompt sensitivity experiments in this thesis. These definitions help standardize the comparison of results from Grounding DINO, PaliGemma2, and Gemma3 models, providing a common framework for analyzing the effects of prompt formulation on performance.

Table 4.12: Definition of Prompt Polarity used in Prompt Sensitivity tests

Prompt Polarity	Explanation
Affirmative	Positive framing that asserts object presence. Example: “Is it a vehicle?” or “The object is a vehicle.”
Negative	Negative framing that denies object presence. Example: “Is it not a vehicle?” or “The object is not a vehicle.”

Table 4.13: Definition of Prompt Types used in Prompt Sensitivity tests

Prompt Type	Explanation
General Category	Basic object category with no extra context. Example: “vehicle” or “car”
Contextual	Object category + environmental or scene details. Example: “car on snow” or “aerial view of car”
Instructional	Clear instruction for filtering. Example: “Only consider car. Ignore shadows. . .”
Instructional + Contextual	Combines both Instructional and Contextual elements in the prompt.

Prompt Sensitivity test on Grounding DINO

The results in Table 4.14 demonstrate that Grounding DINO’s object detection performance is highly sensitive to the formulation of prompts, particularly the prompt type. While all prompts tested were Affirmative in polarity, prompts classified as Contextual (those including environmental or scene details such as "on snow" or "aerial view") consistently achieved higher mAP scores compared to General Category prompts. For instance, the best performing prompt “a car on snow” achieved an mAP of 0.0613 and mAP@50 of 0.1539. In contrast, more generic prompts like “vehicle .” yielded higher mAP@50 (0.1944) but lower mAP (0.0548), suggesting that while general prompts promote higher recall but they often result in less precise bounding boxes.

These findings indicate a tradeoff: Contextual prompts improve detection precision and localization, especially for small or ambiguous objects, while General Category prompts provide broader object coverage at the cost of spatial accuracy. Overall, this emphasizes the importance of careful prompt type selection when configuring Grounding DINO for use in pseudo-labeling pipelines, particularly in domains where object appearance varies widely due to environmental factors.

Table 4.14: Prompt Sensitivity Test on Grounding DINO

Prompt	Polarity	Type	Layer: Base obj detection	
			mAP	mAP@50
a car on snow .	Affirmative	Contextual	0.0613	0.1539
an aerial view of car .	Affirmative	Contextual	0.0557	0.1288
a vehicle in snow .	Affirmative	Contextual	0.0557	0.1284
vehicle .	Affirmative	General Cat- egory	0.0548	0.1944
a small car .	Affirmative	Contextual	0.0489	0.1532
an automobile .	Affirmative	General Cat- egory	0.0447	0.1451
vehicle on snow .	Affirmative	Contextual	0.0444	0.1101
car .	Affirmative	General Cat- egory	0.0437	0.1833
vehicle in snow .	Affirmative	Contextual	0.0436	0.1090
vehicle in snow.	Affirmative	Contextual	0.0436	0.1090
top view of a vehicle from a drone in snowy region.	Affirmative	Contextual	0.0397	0.0821
an automobile from top view .	Affirmative	Contextual	0.0375	0.0879
a vehicle .	Affirmative	General Cat- egory	0.0361	0.1210
top view of a vehicle .	Affirmative	Contextual	0.0344	0.0797

Prompt Sensitivity test on PaliGemma2

The results in Table 4.15 show that PaliGemma2’s semantic filtering performance is highly sensitive to both prompt polarity and prompt type. In particular, negative polarity prompts consistently outperformed affirmative ones across both Pre-Filter and Post-Filter layers. For example, the best-performing prompt, “The object in the image is not a vehicle” achieved the highest Post-Filter mAP 0.0991 and mAP@50 0.2510, outperforming similar affirmative prompts such as “The object in the image is a vehicle”

which achieved Post-Filter mAP 0.0723 and mAP@50 0.1648. This suggests that negative phrasing guides the model toward stricter, more conservative filtering, reducing false positives.

Additionally, prompt type also influenced results. Prompts using broader categories (General Category, e.g., "vehicle") generally performed better than more specific, contextual phrases (e.g., "car on snow"). For instance, the negative general prompt "not a vehicle" outperformed more specific negative contextual prompts such as "not a car on snow" or "not an automobile from top view." These results suggest that broader categories help the model generalize more effectively during semantic filtering.

Overall, the results confirm that both prompt polarity and prompt type must be carefully considered when designing prompts for PaliGemma2 filtering. The strong effect of negative polarity highlights model specific behavior and suggests that optimal filtering may require different strategies for different vision-language models.

Table 4.15: Prompt Sensitivity Test on PaliGemma2

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
The object in the image is not a vehicle. Answer with 'yes' or 'no'.	Negative	General Category	0.0592	0.2051	0.0991	0.2510
The object in the image is not a car. Answer with 'yes' or 'no'.	Negative	General Category	0.0556	0.1935	0.0832	0.2071
The object in the image is a vehicle. Answer with 'yes' or 'no'.	Affirmative	General Category	0.0478	0.1591	0.0723	0.1648

Prompt	Polarity	Type	Layer 2 - Pre-Filter		Layer 4 - Post-Filter	
			mAP	mAP@50	mAP	mAP@50
The object in the image is an automobile from top view. Answer with 'yes' or 'no'.	Affirmative	Contextual	0.0460	0.1508	0.0507	0.1194
The object in the image is an aerial view of car. Answer with 'yes' or 'no'.	Affirmative	Contextual	0.0451	0.1523	0.0016	0.0056
The object in the image is not a car on snow. Answer with 'yes' or 'no'.	Negative	Contextual	0.0445	0.1727	0.0741	0.1912
The object in the image is not an automobile from top view. Answer with 'yes' or 'no'.	Negative	Contextual	0.0444	0.1708	0.0562	0.1535
The object in the image is not a small car. Answer with 'yes' or 'no'.	Negative	Contextual	0.0439	0.1671	0.0721	0.1965
The object in the image is a small car. Answer with 'yes' or 'no'.	Affirmative	Contextual	0.0427	0.1392	0.0483	0.1088

Prompt	Polarity	Type	Layer 2 - Pre-Filter		Layer 4 - Post-Filter	
			mAP	mAP@50	mAP	mAP@50
The object in the image is a car. Answer with 'yes' or 'no'.	Affirmative	General Category	0.0415	0.1336	0.0481	0.1229
The object in the image is not an aerial view of car. Answer with 'yes' or 'no'.	Negative	Contextual	0.0351	0.1269	0.0309	0.0843
The object in the image is a car on snow. Answer with 'yes' or 'no'.	Affirmative	Contextual	0.0106	0.0306	0.0013	0.0033

Prompt Sensitivity Test on Gemma3

The results in Table 4.16 show that Gemma3’s performance is highly sensitive to both prompt polarity and prompt type, though the effects differ from those observed with PaliGemma2. In contrast to PaliGemma2, where negative polarity improved performance, Gemma3 consistently achieved better results with affirmative prompts across both Pre-Filter and Post-Filter layers. For example, the affirmative prompt “Is it an automobile in snow?” produced the highest Post-Filter mAP@50 of 0.1507, while similar negative polarity prompts resulted in much lower scores (e.g., “Is it not a vehicle in snow?” yielded only 0.0150). This highlights that prompt sensitivity behavior varies between models.

Prompt type also had a strong impact on performance. Prompts that combined contextual and instructional elements (e.g., “automobile in snow” or “vehicle in snow. Ignore buildings or shadows...”) consistently outperformed simpler general category prompts. Slight variations in wording resulted in noticeable differences in mAP. For

example, “car in snow” yielded a Post-Filter mAP@50 of 0.1110, “vehicle in snow” reached 0.1496, and “automobile in snow” gave the highest at 0.1507. This suggests that broader terms tend to generalize better, whereas narrow or specific phrases underperform.

Across all prompt types, mAP scores in Layer 4 were consistently higher than those in Layer 2, demonstrating that segmentation based filtering (using object masks) is more effective for refining detections and suppressing noise than bounding box based filtering in Layer 2.

Finally, prompts that included additional image origin cues (e.g., “drone-captured aerial image”) had negligible impact on performance. For example, the prompt “Is it an automobile in snow?” with and without the phrase “drone-captured aerial image” produced nearly identical results (Post-Filter mAP@50 of 0.1507 and 0.1468, respectively), suggesting that Gemma3’s filtering is more influenced by object and scene-related context than by metadata.

Overall, these findings highlight that affirmative polarity, careful word choice, and the use of contextual and instructional phrasing are critical for maximizing filtering effectiveness with Gemma3. Furthermore, optimal prompt engineering strategies must be tailored to each VLM, as behaviors differ significantly across models.

Table 4.16: Prompt Sensitivity Test on Gemma3

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
This is a image crop of a drone-captured aerial image. Is it an automobile in snow? Ignore buildings, houses, shadows or any other objects. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional + Contextual	0.0136	0.0642	0.0574	0.1507
This is a image crop of a drone-captured aerial image. Is it a vehicle in snow? Ignore buildings, houses, shadows or any other objects. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional + Contextual	0.0136	0.0642	0.0570	0.1496

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
This is a image crop of a drone-captured aerial image. Does it contain the top view of a vehicle? Do not consider buildings, houses, shadows or any other objects. Only consider car. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional	0.0136	0.0641	0.0569	0.1491
Is it an automobile in snow? Ignore buildings, houses, shadows or any other objects. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional + Contextual	0.0135	0.0635	0.0562	0.1468

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
This is a image crop of a drone-captured aerial image. Is it a vehicle? Ignore buildings, houses, shadows or any other objects. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional	0.0134	0.0629	0.0558	0.1457
This is a image crop of a drone-captured aerial image. Does it contain the top view of a car? Do not consider buildings, houses, shadows or any other objects. Only consider car. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional	0.0134	0.0642	0.0543	0.1397

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
This is a image crop of a drone-captured aerial image. Does it contain the top view of a vehicle in snow? Do not consider buildings, houses, shadows or any other objects. Only consider car. Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional + Contextual	0.0132	0.0634	0.0515	0.1337
This is a image crop of a drone-captured aerial image. Is it a car in snow? Answer strictly with either 'yes' or 'no' in lowercase.	Affirmative	Instructional + Contextual	0.0118	0.0576	0.0459	0.1110

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
This is a image crop of a drone-captured aerial image. Is it not a car? Answer strictly with either 'yes' or 'no' in lowercase.	Negative	Instructional	0.0074	0.0366	0.0149	0.0572
This is a image crop of a drone-captured aerial image. Is it not a car in snow? Answer strictly with either 'yes' or 'no' in lowercase.	Negative	Instructional + Contextual	0.0028	0.0098	0.0068	0.0248
This is a image crop of a drone-captured aerial image. Is it not a vehicle in snow? Answer strictly with either 'yes' or 'no' in lowercase.	Negative	Instructional + Contextual	0.0021	0.0098	0.0048	0.0150

Prompt	Polarity	Type	Layer: Pre-Filter		Layer: Post-Filter	
			mAP	mAP@50	mAP	mAP@50
This is a image crop of a drone-captured aerial image. Is it not a vehicle? Answer strictly with either 'yes' or 'no' in lowercase.	Negative	Instructional	0.0057	0.0244	0.0027	0.0087

Effect of Prompt Polarity on VLM Performance

The experiments revealed that the effect of prompt polarity in filtering effectiveness of vision-language models, whether a prompt is framed as affirmative or negative, is strongly model dependent.

For PaliGemma2, prompts with negative polarity consistently outperformed their affirmative counterparts. As seen in Table 4.15, the negative prompt "The object in the image is not a vehicle." (General Category) achieved the highest Post-Filter mAP@50 of 0.2510, while the corresponding affirmative prompt "The object in the image is a vehicle." scored only mAP@50 of 0.1648. Similarly, "not a car" (negative, General Category) reached mAP@50 of 0.2071, compared to the affirmative "a car" with just mAP@50 of 0.1229.

Conversely, Gemma3 exhibited the opposite behavior, performing better with affirmative prompts. As shown in Table 4.16, multiple pairs of affirmative versus negative prompts demonstrate this effect. For example, the prompt "Is it a vehicle?" achieved a Post-Filter mAP@50 of 0.1457, while the negative version "Is it not a vehicle?" scored only 0.0087. Similarly, "Is it a vehicle in snow?" yielded 0.1496, compared to "Is it not a vehicle in snow?" at 0.0150. Finally, "Is it a car in snow?" achieved 0.1110, whereas "Is it not a car in snow?" produced only 0.0248. This consistent pattern across all prompt

pairs demonstrates that Gemma3 is more effective when guided by affirmative phrasing, unlike PaliGemma2 where negative prompts improved filtering.

These contrasting results highlight that the impact of prompt polarity is not universal, it is highly dependent on the underlying VLM model. While PaliGemma2 benefits from negatively framed prompts, Gemma3 performs better with affirmative ones. This finding reinforces the importance of model specific prompt engineering when leveraging vision-language models.

4.3.3 Extended Evaluation on 500 Images

To further validate the consistency and scalability of the proposed pipeline, the evaluation was extended to a larger and more diverse test set consisting of 500 images. This set included a mix of images with both moderately sized and small cars, offering a more realistic distribution of object sizes and scenarios. The same pipeline configuration established in earlier tests, including sliding window processing and WBF for duplicate detection removal was retained to ensure comparability. This extended evaluation aimed to assess whether the improvements observed on smaller test sets would generalize to larger data volumes and more varied visual contexts. The results and observations from this test set are detailed below.

Test Set – 500-Image - Result

The results for the extended test on 500 images, shown in Table 4.17, reveal that while the pipeline using PaliGemma2 performs better in the early stage (Layer 2) by retaining more valid detections during pre-filtering, the pipeline with Gemma3 as the filtration model shows superior performance in the later stages particularly after SAM2 based video tracking and the final WBF de-duplication. A significant improvement in performance is observed after SAM2 based video tracking, highlighting its value in recovering missed detections across frames. This shows that model choice at each layer and temporal propagation significantly impacts the pipeline’s effectiveness. Additionally, de-duplicating the propagated masks consistently improves performance, emphasizing its importance in large-scale pseudo-labeling.

Notably, the drop in performance in the early layers on the 500-Image set, especially with Gemma3 during pre and post-filtering, highlights the challenge of prompt sensitivity and model specific tuning. However, the robustness of the pipeline is evident in how later stages (SAM2 video tracking) is able to recover and stabilize the results. This underlines the effectiveness of combining temporal propagation and VLM-guided filtering for consistent pseudo-label generation at larger scales.

Table 4.17: Result at each layer - 500-Image

Layer	Gemma3		PaliGemma2	
	mAP	mAP@50	mAP	mAP@50
Layer 1 - Base Object Detector	0.0415	0.1737	0.0415	0.1737
Layer 2 - Pre-Filter	0.0003	0.0022	0.0479	0.1725
Layer 4 - Post-Filter	0.0005	0.0014	0.0010	0.0024
Layer 5 - Video Tracking	0.0930	0.1953	0.0796	0.1773
Post Processing - De-duplicate using WBF	0.1040	0.2246	0.0945	0.2163

Scalability and Consistency Across Test Sets

A comparison between the 100-Image and 500-Image test results (Tables 4.11 and 4.17) provides valuable insights into the scalability and consistency of the proposed pipeline. Despite the larger scale and increased complexity of the 500-Image test set, the pipeline maintains stable performance characteristics, particularly after the temporal propagation and final WBF de-duplication stages.

For instance, with Gemma3 as the VLM, the final mAP and mAP@50 on the 100-Image set are 0.0883 and 0.2010, respectively, while on the 500-Image set, the corresponding values increase to 0.1040 and 0.2246. Similarly, the pipeline using PaliGemma2 also shows a consistent pattern, improving from 0.0897 / 0.2213 on the smaller test set to 0.0945 / 0.2163 on the larger one. These results suggest that the pipeline not only scales well with larger datasets but also maintains detection quality.

4.4 Key Findings

- Applying a sliding window to images was essential for enabling Grounding DINO to detect very small objects. Without windowing, small cars were consistently missed, resulting in poor detection performance.
- The box threshold in Grounding DINO significantly impacts detection results. Lower thresholds improve recall but introduces more noise.
- Reducing the box threshold increases the number of detections, including many false positives, which leads to higher computational load and longer processing time.
- The text threshold in Grounding DINO did not show any effect on detection performance.
- Weighted Boxes Fusion (WBF) was more effective than Non-Maximum Suppression (NMS) for removing duplicate detections.
- The pipeline achieves better overall performance when using Gemma3 as the vision-language model for filtration compared to PaliGemma2; however, this comes at the cost of significantly increased computation time.
- Incorporating a vision-language model (VLM) for filtering significantly improves the precision of pseudo-label outputs.
- Filtering based on segmentation masks (Layer 4) is more effective than filtering based on bounding boxes (Layer 2), resulting in higher precision and better suppression of noise.
- Prompt formulation significantly impacts detection performance in Grounding DINO, as well as filtering performance in PaliGemma2 and Gemma3.
- Video tracking with SAM2 is a valuable step but must be applied carefully. Propagating only high confidence detections helps prevent the spread of noise across frames, thereby preserving overall performance and improving result quality.

4.5 Code

The code implemented as part of this work is available in Github <https://github.com/devashish-roy/robust-unsupervised-object-detection>.

Chapter 5

Conclusion

5.1 Conclusion

The conclusions of this thesis, closely aligned with the stated objectives, are:

1. The thesis successfully designed and implemented a multi-layer pseudo-labeling framework that integrates zero-shot open-vocabulary detection (Grounding DINO), vision-language models (Gemma3, PaliGemma2), and segmentation based techniques (SAM2), enabling unsupervised annotation from unlabeled image data.
2. The experiments demonstrated that Vision-Language Models significantly improve the semantic filtering of noisy detections. Prompt based reasoning was shown to be highly effective in suppressing false positives and improving the precision of pseudo-labels.
3. The analysis of Grounding DINO configurations highlighted the importance of box threshold tuning. Lower box thresholds were found to improve recall, especially for small objects, while also increasing computational cost, demonstrating a clear trade-off between detection quality and efficiency. The text threshold, however, did not show any significant influence on performance.
4. Spatial techniques such as sliding window proved critical for detecting small and subtle objects, while temporal techniques using SAM2 video tracking improved

recall and temporal consistency across frames. Confidence based thresholding during propagation in video tracking was shown to further enhance final results.

5. The study confirmed that prompt sensitivity significantly affects both object detection (Grounding DINO) and semantic filtering (PaliGemma2 and Gemma3). For Grounding DINO, prompts using contextual descriptions (e.g., “a car on snow”) achieved the best detection precision (highest mAP), while general category prompts (e.g., “vehicle .”) achieved the highest mAP@50 but lower mAP, promoting higher recall at the expense of bounding box precision.

For PaliGemma2, negative polarity prompts performed consistently better than affirmative ones. The best results were achieved with the broad, negatively framed prompt “not a vehicle,” which significantly outperformed narrower and affirmative variants. This confirms that negative polarity combined with broad prompt wording encourages stricter semantic filtering in PaliGemma2.

Conversely, Gemma3 demonstrated better performance with affirmative polarity prompts, highlighting that prompt polarity effects are strongly model-dependent. Broader category prompts such as “vehicle” or “automobile” outperformed narrower ones like “car.” In addition, Instructional + Contextual prompts further enhanced performance.

Overall, these results emphasize that prompt sensitivity is highly model specific across both object detection and semantic filtering stages. Optimizing prompt formulation, in terms of polarity, category breadth, and contextual cues is essential for maximizing performance in pseudo-labeling pipelines.

In conclusion, this work demonstrates that combining zero-shot, open-vocabulary detection, VLM-guided filtering, prompt engineering, segmentation, and video tracking can produce a scalable multi-layered unsupervised pipeline for annotation. The proposed framework lays the groundwork for future enhancements, as outlined in the Future Work subsection (Section 5.3).

5.2 Limitation

Despite the framework’s promising results, several limitations remain:

1. The performance of both Grounding DINO and vision-language models (Gemma3, PaliGemma2) is highly sensitive to prompt phrasing and formulation. Small changes in wording can cause large variations in detection or filtering outcomes. This makes the pipeline dependent on careful manual prompt engineering, which may limit its generalization to other object categories or domains without additional tuning.
2. Evaluation was limited to the Nordic Vehicle Dataset (NVD). Generalization to other object categories or domains (e.g., urban or maritime) is yet to be tested.
3. SAM2 video tracking, while effective for improving temporal consistency, significantly increases computation cost.
4. For SAM2 Video tracking to be effective, it is critical that noisy detections are filtered out in earlier layers; otherwise, noise will propagate and degrade final output quality.

5.3 Future Work

While the proposed framework demonstrates promising results, several directions can be explored to further improve its effectiveness and scalability.

The current evaluation was conducted on a single dataset (NVD). Evaluating the pipeline on a larger and more diverse set would help assess its generalization across object types, scenes, and domains. In particular, validating this framework on widely adopted aerial object detection datasets such as DOTA [42] would provide insight into its performance in complex, high-density, and multi-scale environments.

Another promising direction is to enhance the performance of the base detector, which is a critical component in the pipeline. This could involve experimenting with alternative prompt formulations for Grounding DINO or replacing it with other zero-shot, open-vocabulary detectors such as OWL-ViT [43] or GLIP [32]. Improving the initial

detection stage can significantly impact recall, especially for small or occluded objects, and thus improve the effectiveness of the entire pseudo-labeling framework.

The modular architecture of the framework presents further opportunities for expansion. Currently, only the vision-language model (e.g., Gemma3 or PaliGemma2) is configurable. Extending this flexibility to include customizable modules for detection, filtering, and segmentation would allow adaptation to a wider range of computational constraints and use cases.

Finally, incorporating alternative vision-language models such as LLaVA [44] could be explored to further enhance semantic filtering capabilities.

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